

Primary Exploration of Python Packages Optiland and xrt for Ray Optics.

August 18, 2026

1 Division of software responsibilities.

Optiland is used for sequential visible-light surfaces, glass dispersion, reflection, and image-plane tracing. xrt/raycing is used for the geometric source-mirror-screen beamline, polarization weights, finite apertures, and X-ray material reflectivity. In the lens notebook, xrt's documented external refractive-index interface is used as a material bridge. Native xrt refractive elements are aimed primarily at compound refractive X-ray lenses, so ordinary visible spherical lenses are traced by Optiland rather than being forced into an inappropriate X-ray model.

2 Plane-mirror simulation

2.1 Default configuration

The supplied example is an opaque gold mirror at a wavelength of 0.154 06 nm. The equivalent photon energy is

$$E_\gamma = \frac{hc}{\lambda} = \frac{1239.841\,984\,\text{eV nm}}{0.154\,06\,\text{nm}} \simeq 8.048\,\text{keV}. \quad (1)$$

This default lies in the intended range of the xrt scattering-factor tables. For a visible or infrared mirror, the notebook requires the constant mode and the measured coating values of R and A .

Table 1: Principal default parameters in `MirrorConfig`.

Parameter	Default	Role
Wavelength	0.154 06 nm	photon energy/material response
Grazing angle	0.50°	angle from the xrt mirror surface
Source-mirror distance	1000 mm	incoming propagation
Mirror-screen distance	800 mm	outgoing propagation
Source size (x, z)	0.40 mm $\times$ 0.20 mm	footprint size
Divergence (x, z)	(0.10, 0.05) mrad	angular broadening
Mirror material	Au, 19.32 g/cm ³	xrt optical constants
Rays	20 000	Monte-Carlo sample size

2.2 Geometry and reflection invariant

For incident unit direction $\hat{\mathbf{d}}$ and unit surface normal $\hat{\mathbf{n}}$, the reflected direction is

$$\hat{\mathbf{d}}_r = \hat{\mathbf{d}} - 2(\hat{\mathbf{d}} \cdot \hat{\mathbf{n}})\hat{\mathbf{n}}. \quad (2)$$

Thus a mirror rotation by $\Delta\theta$ rotates the specular beam by $2\Delta\theta$. In the default xrt layout the expected vertical deflection is therefore 1.00°. The screen centre is placed on this direction, so its

elevation relative to the mirror is

$$z_s = L \sin(2\theta) = 800 \text{ mm} \sin(1.00^\circ) \simeq 13.96 \text{ mm}. \quad (3)$$

The notebook asserts the unit length of the reflected direction and checks the mean Monte-Carlo deflection against 2θ .

The Optiland illustration uses the same vector law in a visually convenient folded geometry: a 45° plane mirror turns a beam initially along $+z$ into $+y$. Its detector is positioned normal to that reflected branch.

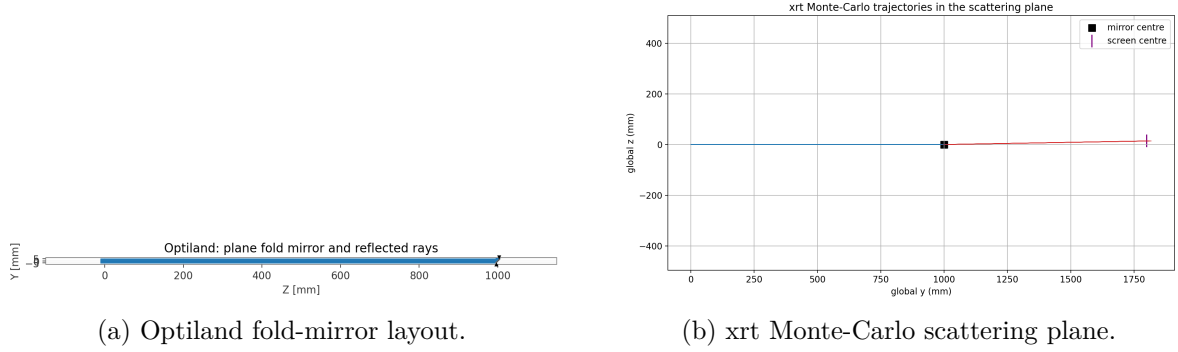


Figure 1: The two geometric views implement the same specular-reflection law but use different coordinate conventions.

2.3 Reflectance, absorptance, and detector distribution

xrt obtains the complex s - and p -polarized amplitudes from the selected material and incidence angle. The intensity reflectances are

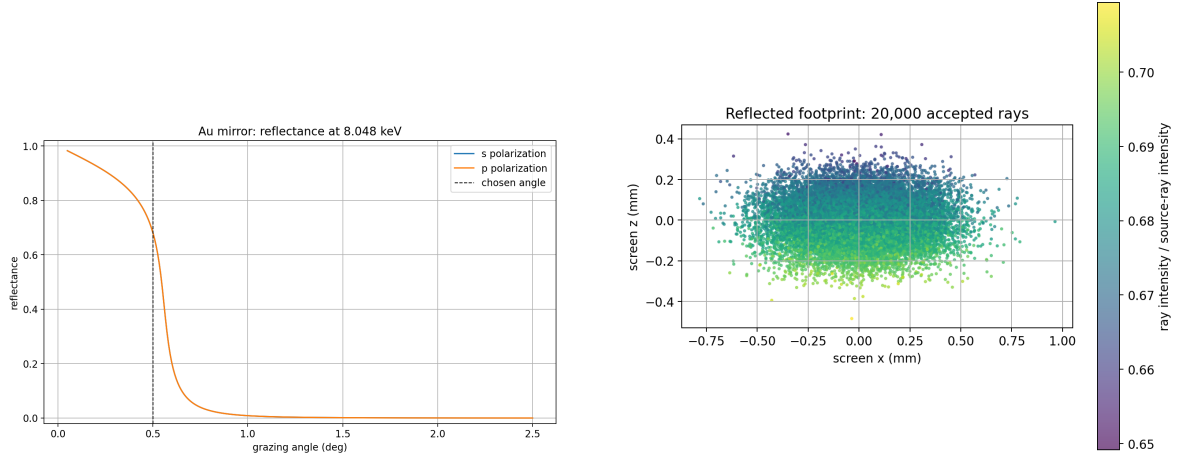
$$R_s = |r_s|^2, \quad R_p = |r_p|^2. \quad (4)$$

These weights are applied to the ray coherency matrix. For an opaque thick mirror, transmission is neglected and

$$A = 1 - R, \quad T = 0, \quad R + A + T = 1. \quad (5)$$

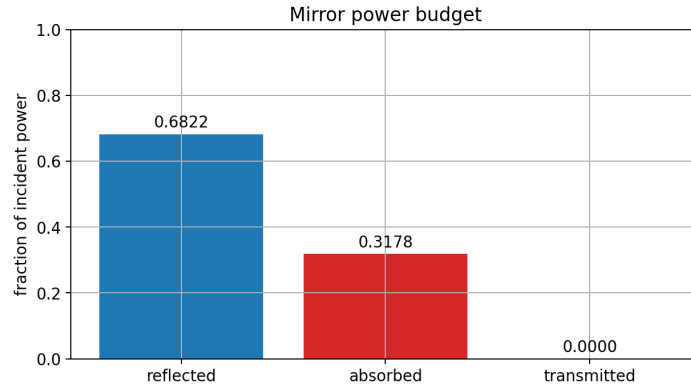
In the user-defined constant mode, all three branches are retained in the reported budget, with $T = 1 - R - A$. xrt traces the reflected branch geometrically; the table and bar chart account for the non-reflected power.

Figure 2 should be interpreted as follows. The reflectance plot shows the rapid angle and polarization dependence expected of an X-ray mirror. The vertical dashed line marks the selected grazing angle. The detector distribution convolves source size and divergence and is clipped by the finite mirror. Increasing either divergence broadens the corresponding screen coordinate approximately linearly with propagation distance; decreasing the mirror width or length lowers the accepted-ray fraction.



(a) Polarization-dependent material reflectance.

(b) Reflected screen footprint.



(c) Reflected, absorbed, and transmitted fractions.

Figure 2: Plane-mirror outputs generated by `Plane Mirror.ipynb`.

3 Movable lens-group simulation

3.1 Prescription and exact model

The default three-element group is summarized in Table 2. The front and rear radii, thickness, glass, position, wavelengths, field angles, aperture, and fixed sensor location are all editable. Removing either the second or third tuple produces a two-lens model without changing the builder.

Table 2: Default front-vertex prescription and paraxial reference values.

Lens	z (mm)	R_1 (mm)	R_2 (mm)	d (mm)	Glass
L1	0	100	-100	5	N-BK7
L2	45	150	-150	4	N-SF5
L3	70	120	-120	5	N-BK7

For the first-order comparison, each element focal length is estimated with the thick lensmaker equation in air,

$$\frac{1}{f} = (n - 1) \left[\frac{1}{R_1} - \frac{1}{R_2} + \frac{(n - 1)d}{nR_1R_2} \right]. \quad (6)$$

The reference indices in the notebook give approximately $f_1 = 97.58$ mm, $f_2 = 112.09$ mm, and $f_3 = 116.93$ mm. Optiland does not use these constants for the exact trace: it loads the named

glass and evaluates its dispersion at each of 486.1 nm, 587.6 nm, and 656.3 nm.

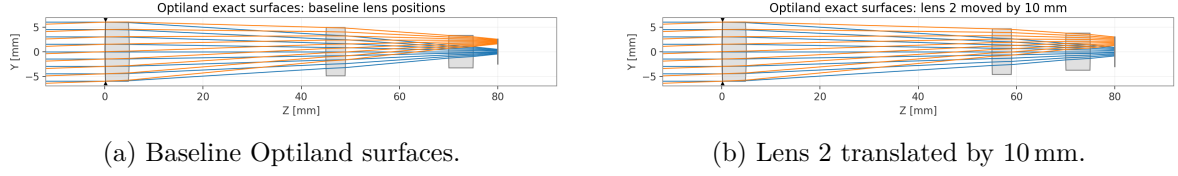


Figure 3: Exact thick-lens layouts at the same fixed sensor plane.

3.2 Why lens translation moves the focus

In one meridional plane, free propagation by distance d and refraction by an ideal thin lens of focal length f are represented by

$$P(d) = \begin{pmatrix} 1 & d \\ 0 & 1 \end{pmatrix}, \quad L(f) = \begin{pmatrix} 1 & 0 \\ -1/f & 1 \end{pmatrix}. \quad (7)$$

The group matrix contains alternating P and L factors. Moving lens 2 changes both adjacent propagation distances, so the group power and the principal-plane locations change even though the lens curvature does not. For collimated input, a matrix $M = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$ places the paraxial focus a distance $-A/C$ after the output reference plane.

With the default reference indices, the paraxial calculation predicts a focus near $z = 79.88$ mm, close to the fixed sensor at 80 mm. Translating L2 from $z = 45$ mm to $z = 55$ mm moves the first-order focus to about $z = 83.96$ mm. Therefore the moved configuration should have a larger spot at the unchanged sensor. Exact numerical values in the Optiland table can differ because the spherical surfaces include thickness, aberration, field angle, vignetting, and wavelength-dependent glass data.

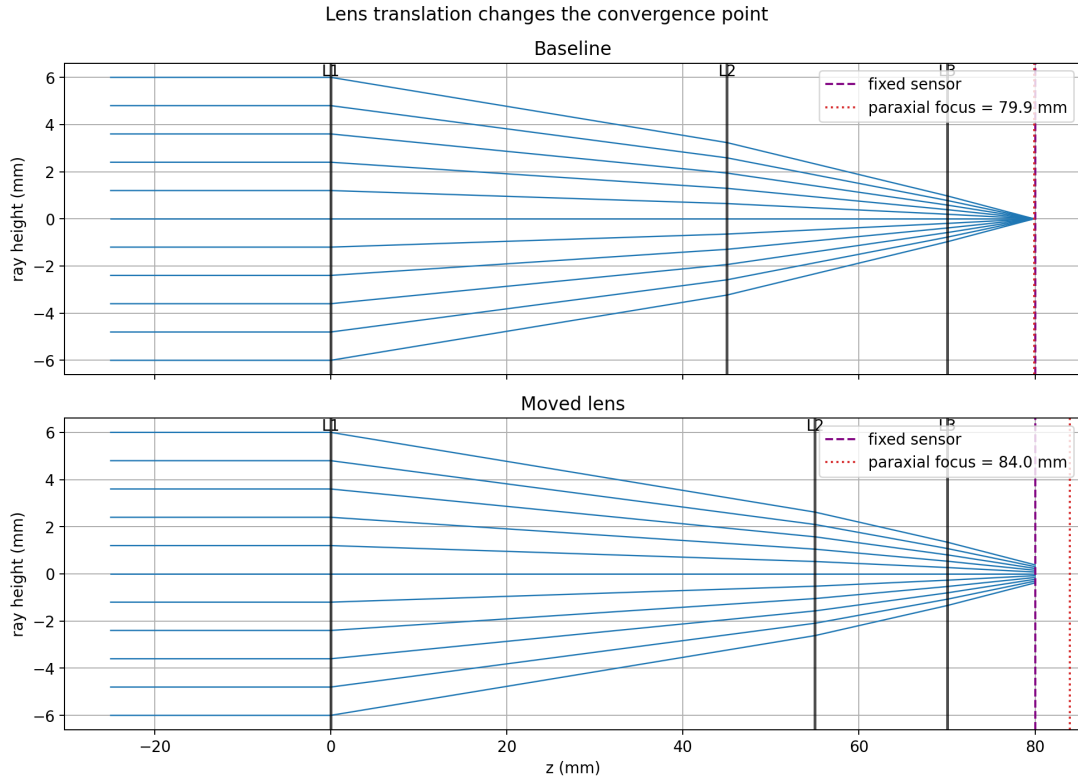


Figure 4: Independent paraxial trajectories before and after translating L2.

3.3 Spot diagrams and position sensitivity

The image-plane RMS spot radius is calculated from accepted rays as

$$r_{\text{RMS}} = \sqrt{\langle (x - \bar{x})^2 + (y - \bar{y})^2 \rangle}. \quad (8)$$

A smaller value at the fixed sensor means better convergence there, not necessarily a globally optimized imaging system. Differences between the three wavelength panels show chromatic residuals. At the off-axis field, coma, astigmatism, and field curvature may also contribute.

The two-dimensional sweep moves L2 and L3 while keeping L1 and the sensor fixed. A dark valley indicates combinations that make the first-order sensor height small. The sweep is an initializer: promising positions must be copied back to the prescription and verified with the exact Optiland trace. Invalid overlapping prescriptions are masked rather than silently traced.

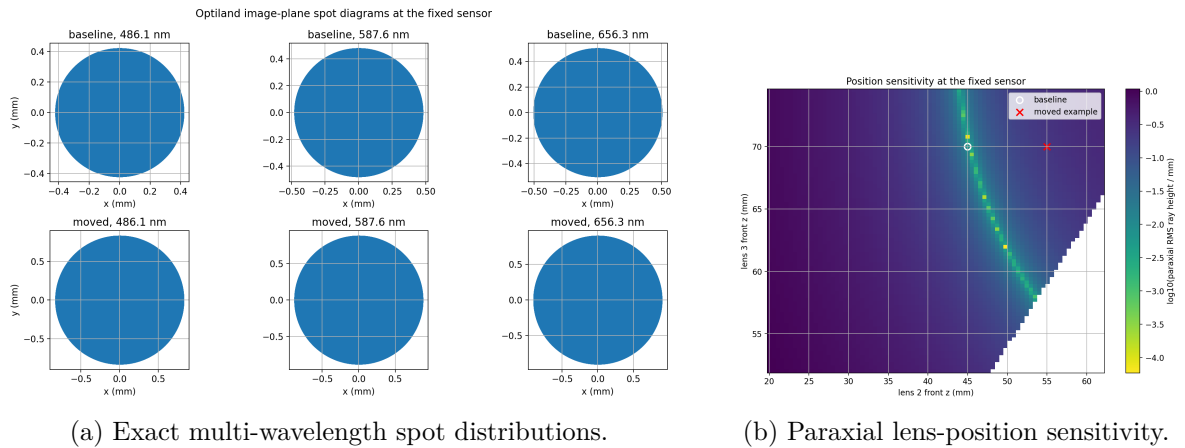


Figure 5: Fixed-sensor convergence metrics from `lens_group.ipynb`.

4 Validation, interpretation, and limitations

The notebooks use several checks that are independent of a visually plausible plot:

1. Equation (2) predicts the 2θ mirror deflection, which is compared with the mean xrt ray direction.
2. Equation (5) is asserted numerically, including a possible transmitted fraction in constant mode.
3. The external visible refractive indices returned by xrt are checked against the requested constants.
4. The paraxial lens calculation predicts the direction of focus shift; the exact Optiland RMS spot verifies its consequence at the fixed sensor.
5. Lens overlap and sensor-order checks reject non-physical position combinations during the sweep.

The simulations are geometrical-optics models. They do not by themselves include diffraction-limited point-spread functions, mirror roughness scatter, surface figure error, mechanical tolerances, thermal deformation, ghost paths, or a measured coating spectrum. The xrt material tables are suitable for the X-ray/EUV example, not for an arbitrary visible mirror. Conversely, visible glass catalogue data should not be interpreted as an X-ray compound refractive lens model. For an experiment, measured source phase space, coating data, surface prescriptions, and detector response should replace the defaults.

5 Recommended parameter studies

For the mirror, sweep wavelength and grazing angle together, plot R_s and R_p , and record accepted flux and screen RMS width. This separates a material-response loss from aperture clipping. For the lens group, first use the fast position heat map to select a region, then evaluate the exact multi-wavelength RMS spot and off-axis fields at each candidate. A final design should optimize radii, spacings, glass choices, and sensor position jointly rather than interpreting one translated configuration as an optimum.

References

- [1] Optiland documentation, “Quickstart—Your First 5 Minutes” and reflective examples. <https://optiland.readthedocs.io/>
- [2] xrt documentation, “Raycing backend” and “Using xrt as X-ray calculator.” <https://xrt.readthedocs.io/>